

PHISHING EMAIL DETECTION & AWARENESS SYSTEM

INTRODUCTION

Phishing is one of the most common cyber threats faced by individuals and organizations worldwide. It is a type of social engineering attack in which cybercriminals send fraudulent emails that appear to come from trusted organizations, such as banks, online services, or well-known companies. The primary objective of phishing attacks is to trick users into revealing sensitive information, including usernames, passwords, banking details, credit card information, and one-time passwords (OTPs).

Phishing emails often create a sense of urgency, fear, or excitement to manipulate recipients into taking immediate action. Attackers may use fake websites, malicious links, or deceptive attachments to steal information or install malware on a victim's device. As phishing techniques continue to evolve, users must be able to recognize suspicious emails and understand the indicators of phishing attempts.

This project focuses on analyzing phishing email samples to identify common phishing characteristics, evaluate their risk levels, and understand how such attacks operate. The findings from this analysis are used to develop awareness guidelines and preventive measures that can help users protect themselves from phishing attacks and improve overall cybersecurity awareness.

OBJECTIVES

The main objective of this project is to analyze phishing email samples and identify the techniques used by attackers to deceive users. This project aims to improve awareness of phishing attacks and help users recognize suspicious emails before becoming victims.

Specific objectives of this project include:

- To collect and examine phishing email samples.
- To identify common phishing indicators such as suspicious links, fake sender addresses, urgency tactics, and generic greetings.
- To analyze email headers and domains for signs of malicious activity.
- To classify emails based on their risk level (Safe, Suspicious, or Phishing).
- To understand how phishing attacks are carried out and their potential impact on users and organizations.

- To develop awareness guidelines and best practices for preventing phishing attacks.
- To educate users on safe email handling and cybersecurity practices.

SAMPLE 1 ANALYSIS

Email Content:

Subject: Urgent: Your Account Will Be Locked

Dear User,

We detected unusual activity on your account.

To prevent account suspension, please verify your account immediately by clicking the link below:

<http://secure-account-verify.com>

Failure to verify within 24 hours will result in permanent account lock.

Regards,

Security Team

Phishing Indicators Found

- Generic greeting
- Suspicious link
- Urgency language
- Threat of account suspension

Risk Classification:

PHISHING

SAMPLE 2 ANALYSIS

Email Content:

Subject: PayPal Security Alert

Dear Customer,

Your PayPal account has been temporarily restricted due to suspicious login attempts.

Please confirm your identity immediately:

<http://paypal-security-check.net>

If you fail to verify your information within 12 hours, your account will be permanently suspended.

Thank you,

PayPal Support Team.

Phishing Indicators Found

- Fake sender domain
- Generic greeting
- Urgency tactics
- Suspicious URL

Risk Classification:

PHISHING

SAMPLE 3 ANALYSIS

Email Content:

Subject: Congratulations! You Have Won a Gift Card

Hello,

Congratulations! You have been selected as the winner of a ₹25,000 Amazon Gift Card.

Claim your reward now by clicking the link below:

<http://amazon-gift-rewards.xyz>

This offer expires today.

Best Regards,

Rewards Department.

Phishing Indicators Found

- Fake reward offer
- Suspicious URL
- Generic greeting
- Urgency tactics

Risk Classification:

PHISHING

RISK CLASSIFICATION TABLE

Email Sample	Indicators Found	Risk Classification
Sample 1	Suspicious URL, Urgency, Generic Greeting	Phishing
Sample 2	Fake Domain, Urgency, Generic Greeting	Phishing
Sample 3	Fake Reward, Suspicious URL, Urgency	Phishing

PHISHING INDICATORS TABLE

Indicator	Description
Generic Greeting	No recipient name
Suspicious Link	Unknown or fake URL
Urgency	Creates panic
Fake Domain	Pretends to be legitimate

EMAIL HEADER ANALYSIS

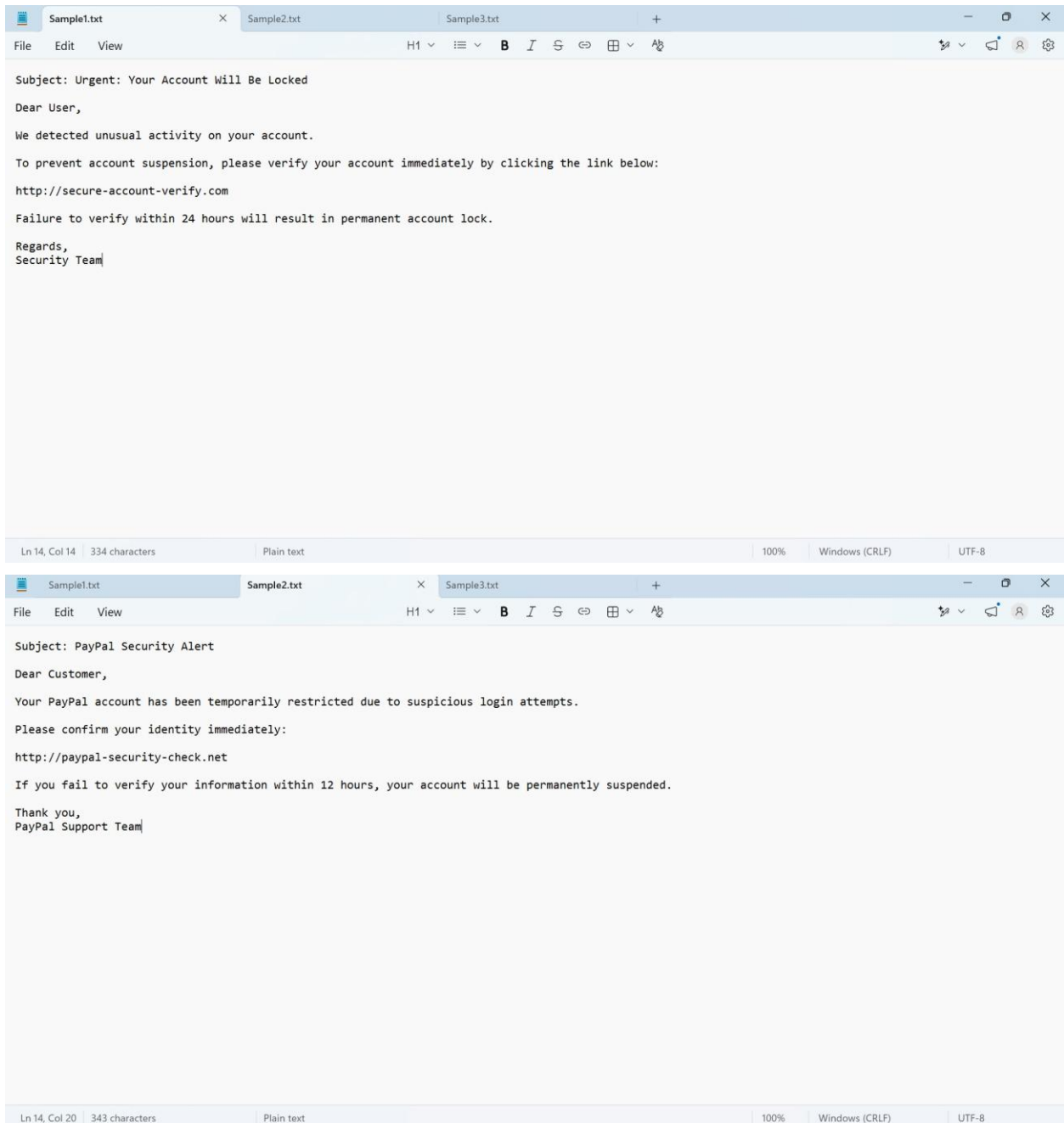
Email headers contain technical information regarding the sender, routing path, and authentication status of an email.

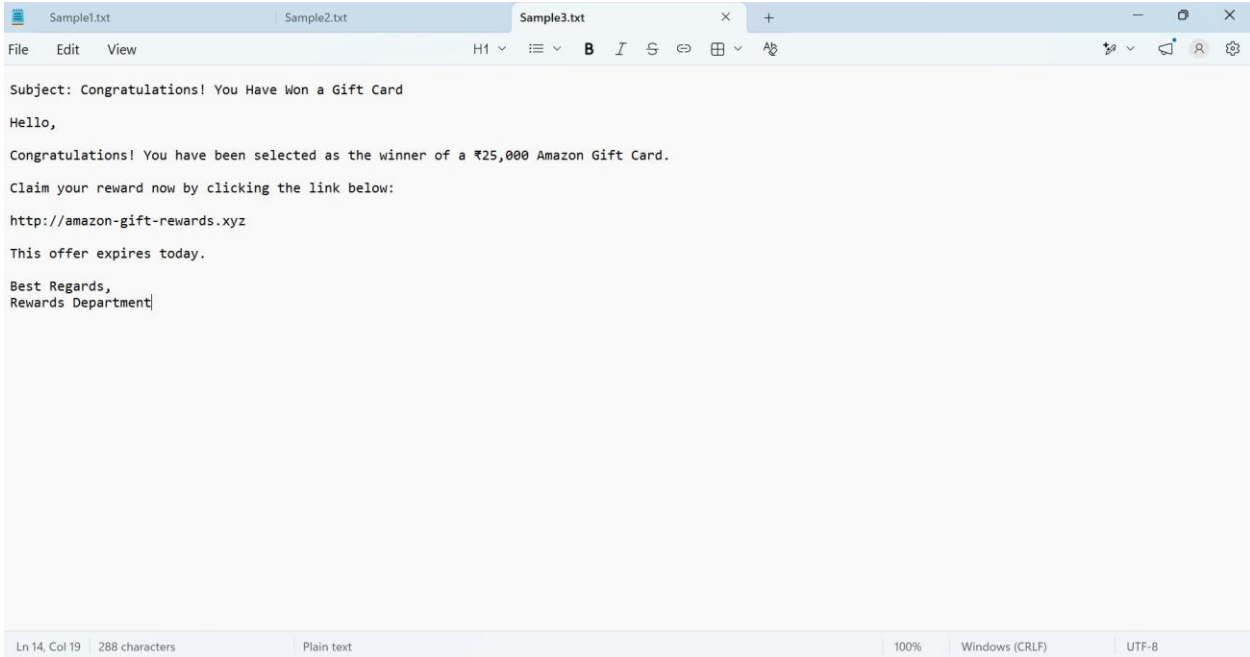
Header analysis helps identify phishing emails by examining SPF, DKIM, and DMARC records.

DOMAIN ANALYSIS

Phishing attackers often use deceptive domains that resemble legitimate organizations.

Suspicious domains were identified in all analyzed samples.





HOW PHISHING ATTACKS WORK

1. Attacker creates fake email.
2. Victim receives email.
3. Victim clicks malicious link.
4. Victim enters credentials.
5. Credentials are stolen.
6. Account becomes compromised.

PREVENTION GUIDELINES

- Verify sender addresses.
- Check URLs carefully.
- Never share passwords.
- Never share OTPs.

- Enable Multi-Factor Authentication.
- Report suspicious emails.

DO'S

- ✓ Verify sender email.
- ✓ Check domain names.
- ✓ Use MFA.
- ✓ Report phishing emails.

DON'TS

- ✗ Click suspicious links.
- ✗ Download unknown attachments.
- ✗ Share passwords.
- ✗ Share OTPs.

CONCLUSION

This project successfully analyzed three phishing email samples and identified common phishing indicators such as suspicious links, generic greetings, urgency tactics, and fake reward offers. Based on these indicators, the emails were classified as phishing emails.

The analysis demonstrated how cybercriminals use deceptive techniques to trick users into revealing sensitive information. By understanding these phishing characteristics, users can better recognize and avoid phishing attacks.

The project also highlighted the importance of email awareness, careful link verification, and following cybersecurity best practices. Implementing the recommended prevention measures can help reduce the risk of phishing attacks and improve overall security awareness.

